IGCSE O-LEVEL
MATHEMATICS

ALGEBRA
NOTES

A Comprehensive Guide for Edexcel & Cambridge

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13 COMPREHENSIVE TOPICS —
STEP-BY-STEP EXAMPLES —
EXAM-FOCUSED

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Complete Guide to IGCSE Algebra Topics

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1 Algebra Toolkit

A strong foundation in algebra begins with understanding its basic building blocks. This section introduces the fundamental concepts that will be used throughout your IGCSE O-Level studies.

1.1 What is a Variable?

Definition: Variable

A **variable** is a symbol (usually a letter such as x, y, or n) that represents an unknown or changing quantity in mathematics. Variables allow us to write general rules and formulas that work for many different values.

Variables are essential because they let us:

- Represent unknown numbers in equations ($x + 3 = 7$)
- Express mathematical relationships ($A = \pi r^2$)
- Describe patterns and rules ($y = 2x + 1$)

1.2 The Concept of Substitution

Definition: Substitution

Substitution is the process of replacing a variable with a specific number to evaluate an expression or check a solution.

Example 1: Simple Substitution

Evaluate $3x + 5$ when $x = 4$:

$$\begin{aligned} 3x + 5 &= 3(4) + 5 \\ &= 12 + 5 \\ &= 17 \end{aligned}$$

Example 2: Checking Solutions

Is $x = 2$ a solution to $4x - 3 = 5$?

$$\begin{aligned} 4(2) - 3 &= 8 - 3 \\ &= 5 \end{aligned}$$

Since $5 = 5$, **yes**, $x = 2$ is a solution.

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1.3 Order of Operations

When working with algebraic expressions, always follow **BIDMAS/PEMDAS**:

1. **B**rackets / **P**arentheses
2. **I**ndices / **E**xponents
3. **D**ivision and **M**ultiplication (left to right)
4. **A**ddition and **S**ubtraction (left to right)

Example 3: Order of Operations

Simplify: $2(3+4)^2 - 5 \times 3$

$$
\begin{aligned}
&= 2(7)^2 - 5 \times 3 && \text{(Brackets first)} \\
&= 2(49) - 5 \times 3 && \text{(Indices)} \\
&= 98 - 15 && \text{(Multiplication)} \\
&= 83 && \text{(Subtraction)}
\end{aligned}
$$

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2 Algebraic Roots & Indices

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1 The Core Laws of Indices

These rules are the foundation for everything involving powers.

Summary of Index Laws

Law	Rule	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$x^2 \times x^5 = x^{2+5} = x^7$
Division	$\dfrac{a^m}{a^n} = a^{m-n}$	$\frac{y^9}{y^4} = y^{9-4} = y^5$
Power of a Power	$(a^m)^n = a^{mn}$	$(k^3)^2 = k^{3\times 2} = k^6$
Power of a Product	$(ab)^n = a^n b^n$	$(3x^2y^4)^3 = 3^3(x^2)^3(y^4)^3 = 27x^6y^{12}$

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2 Special Indices: Zero, Negative, and Fractional

Anything to the power of zero is 1. $a^0 = 1$

- Example: $(3x^2y)^0 = 1$. It doesn't matter how complex the base is.

A negative power means "reciprocal" or "flip it".

$$a^{-n} = \frac{1}{a^n}$$

- Example: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$.
- Example with fractions: $(\frac{x}{y})^{-2} = (\frac{y}{x})^2 = \frac{y^2}{x^2}$.

Fractional powers mean roots. The denominator of the fraction is the type of root. $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

- Example (simple root): $9^{\frac{1}{2}} = \sqrt[2]{9} = 3$.
- Example (harder): $8^{\frac{2}{3}} = (\sqrt[3]{8})^2 = (2)^2 = 4$.

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3 Solving Equations with Indices

This is a very common exam question that combines all the rules.

The Golden Rule of Solving Index Equations

The Goal: Manipulate the equation until you have a **single power with the same base** on both sides.

If $a^x = a^y$, then it must be true that $x = y$.

1. **Unify the Base (Prime Factors):** This is the most general method that solves any question. For every number in the equation, **find its prime factors**. This will reveal the common base you must use. Rewrite all numbers as powers of that single base.
2. **Use the Index Laws** to simplify each side into a single power.
3. **Equate the Powers** and solve the resulting (usually linear) equation.

Problem: Solve for k in the equation $4^{k+3} = 16 \times 2^k$.

1. **Unify the Base** The numbers are 4 and 16. Analyzing their prime factors ($4 = 2^2$ and $16 = 2^4$) shows that the common base is 2.
   - $4 = 2^2$
   - $16 = 2^4$

   The equation becomes: $(2^2)^{k+3} = 2^4 \times 2^k$.

2. **Simplify Each Side**
   - Left Side (Power of a Power): $(2^2)^{k+3} = 2^{2(k+3)} = 2^{2k+6}$.
   - Right Side (Multiplication): $2^4 \times 2^k = 2^{4+k}$.

   Now the equation is: $2^{2k+6} = 2^{4+k}$.

3. **Equate Powers and Solve** The bases are the same, so the powers must be equal.

$$2k + 6 = 4 + k$$
$$2k - k = 4 - 6$$
$$k = -2$$

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✔ **Answer:** The value of k is $\mathbf{-2}$.

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3 Expanding Brackets

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1 Expanding Single Brackets

This is the first step. To expand a single bracket, you **multiply** the term on the outside of the bracket by **every** term on the inside.

- For example, $3x(x+2)$ becomes $3x \times x + 3x \times 2$, which simplifies to $3x^2 + 6x$.
- Be careful with minus signs! For example, $-7x(4-5y)$ becomes $(-7x) \times 4 + (-7x) \times (-5y) = -28x + 35xy$.

Expanding and Simplifying

Sometimes, you have two separate brackets to expand. The goal is to expand each one first, and then **simplify**.

1. **Expand Separately:** Expand each bracket on its own. For example, in $4(x+7) + 5x(3-x)$, first get $4x + 28$ and $15x - 5x^2$.
2. **Collect Like Terms:** "Simplify" means putting similar items together. You combine terms that have the exact same variables and powers.
   - In our example, we have $4x + 28 + 15x - 5x^2$.
   - The "like terms" are $4x$ and $15x$. We can add them to get $19x$.
   - The final simplified answer is $19x + 28 - 5x^2$.

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2 Expanding Double Brackets

To expand double brackets, like $(2x-3)(x+4)$, you must multiply **every term** in the first bracket by **every term** in the second one. A great way to remember this is **FOIL**.

F - First Multiply the first terms in each bracket.

$(\mathbf{2x}-3)(\mathbf{x}+4) \to 2x^2$

O - Outer Multiply the outermost terms.

$(\mathbf{2x}-3)(x+\mathbf{4}) \to +8x$

I - Inner Multiply the innermost terms.

$(2x-\mathbf{3})(\mathbf{x}+4) \to -3x$

L - Last Multiply the last terms in each bracket.

$(2x-\mathbf{3})(x+\mathbf{4}) \to -12$

Finally, collect the like terms: $2x^2+8x-3x-12 = 2x^2+5x-12$.

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3 Special Cases You Must Know

These are shortcuts. If you can spot them, you can solve problems much faster.

Perfect Square

This happens when you square a bracket, like $(x+3)^2$. It means $(x+3)(x+3)$.

- **The Rule:** $(a+b)^2 = a^2 + 2ab + b^2$.
- **How it works:** (Square the first term) + (2 × first term × second term) + (Square the last term).
- **Example:** $(2x+3)^2 = (2x)^2 + 2(2x)(3) + (3)^2 = 4x^2 + 12x + 9$.
- **Warning!** A common mistake is to think $(a+b)^2 = a^2 + b^2$. This is **wrong**!

Difference of Two Squares

This is the most important shortcut! You see it when the brackets are identical, but have opposite signs, like $(a+b)(a-b)$.

- **The Rule:** $(a+b)(a-b) = a^2 - b^2$.
- **Why it works:** The middle 'OI' terms in FOIL always cancel out! $(a+b)(a-b) = a^2 - ab + ab - b^2 = a^2 - b^2$.
- **Be Quick to Spot This!** You will see it a lot, especially with surds. For example, when multiplying conjugates like $(\sqrt{a}+\sqrt{4b})(\sqrt{a}-2\sqrt{b})$, if you notice that $\sqrt{4b} = 2\sqrt{b}$, this is a difference of two squares! The answer is simply $(\sqrt{a})^2 - (2\sqrt{b})^2 = a - 4b$.

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4 Expanding Three Brackets

How to Expand Three Brackets

This is a multi-step process, but it's simple if you stay organized. To expand $(x+1)(x+2)(x+3)$:

1. **Pick two brackets and expand them.** Let's pick $(x+2)(x+3)$. Using FOIL, this becomes $x^2+3x+2x+6 = x^2+5x+6$.
2. **Put the result in a new bracket.** Now our problem is $(x+1)(x^2+5x+6)$.
3. **Multiply everything from the first bracket by the second.**
   - Multiply x by everything in the second bracket: $x(x^2+5x+6) = x^3+5x^2+6x$.
   - Multiply $+1$ by everything in the second bracket: $1(x^2+5x+6) = x^2+5x+6$.
4. **Collect all the like terms.**

$$(x^3+5x^2+6x)+(x^2+5x+6) = \mathbf{x^3+6x^2+11x+6}$$

Exam Tip!

Expanding three brackets is a very important skill and comes up a lot in past papers! Make sure you practice this method until it becomes easy. It is a guaranteed way to score marks if you are careful and organized.

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4 Factorization

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1 Factorising Quadratics (Three Terms)

For expressions with three terms like $ax^2 + bx + c$, we use the **CC Rule**.

The CC Rule: Calculator & Check

1. **C1 - Calculator:** Use your calculator to solve the quadratic equation.
   - Go to the equation mode (e.g., MODE 5-3, or MENU → Equation/Func → Polynomial).
   - Enter the coefficients A, B, and C.
   - The calculator gives you the roots, x_1 and x_2. Now, turn these roots back into factors.

Converting Roots to Factors

- **If the root is an integer:** Change its sign. $x = 5 \implies (x - 5)$
- **If the root is a fraction:** The denominator becomes the coefficient of x, and you change the sign of the numerator. $x = \frac{3}{2} \implies (2x - 3)$

2. **C2 - Check:** This step is crucial. Multiply the first term of each new factor. Does it match the first term of the original expression?
   - **If it matches**, you are done!
   - **If it doesn't match**, find the number you need to multiply by, and place it at the front.

Factorise $4x^2 - 10x + 6$.

1. **Calculator:** Input A=4, B=-10, C=6. The calculator gives roots $x_1 = \frac{3}{2}$ and $x_2 = 1$.
2. **Convert to factors:** This gives you $(2x - 3)$ and $(x - 1)$.
3. **Check:** Multiply the first terms: $(2x)(x) = 2x^2$. This **does not** match the original $4x^2$.

4. **Correct:** We need to multiply $2x^2$ by 2 to get $4x^2$. So, we put a 2 at the front.
5. **Final Answer:** $2(2x-3)(x-1)$.

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2 Factorising Expressions with Two Terms

When you see two terms, your thought process should always be: 1. HCF, then 2. DOTS.

Two-Term Factorising Method

Step 1: Look for a Highest Common Factor (HCF)

Always start here. An HCF can be a number, a letter, or both.

- To factorise $3xy^2 - 6xy$:
- **Numbers:** The HCF of 3 and 6 is 3.
- **Letters:** The HCF of x and x is x. The HCF of y^2 and y is y.
- The overall HCF is $3xy$.
- To find what's inside the bracket, you divide: $\frac{3xy^2}{3xy} = y$ and $\frac{-6xy}{3xy} = -2$.
- **Answer:** $3xy(y-2)$.

Step 2: Check for a Difference of Two Squares (DOTS)

After taking out an HCF (or if there isn't one), check if the remaining expression is a DOTS.

- **Conditions for DOTS:** 1. There is a minus sign. 2. You can square root the first term. 3. You can square root the second term.
- The rule is: $a^2 - b^2 = (a+b)(a-b)$.
- To factorise $50g^2 - 18$:
- First, HCF is 2 $\implies 2(25g^2 - 9)$.
- Now look inside the bracket. It's a DOTS! $\sqrt{25g^2} = 5g$ and $\sqrt{9} = 3$.
- **Answer:** $2(5g+3)(5g-3)$.

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3 Factorising by Grouping (Four Terms)

When you see four terms, you should immediately think of **factorising by grouping**.

How to Factorise by Grouping

1. **Group into pairs.** Group the first two terms and the last two terms.
2. **Factorise each pair.** Find the HCF for each pair separately.
3. **Factor out the common bracket.** After step 2, you will have a bracket that is common to both terms. Factor this bracket out.

Factorise $ab + 3b + 2a + 6$.

- **Step 1 & 2:**
  - For the first pair, $ab + 3b$, the HCF is b. Factoring it out gives $b(a + 3)$.
  - For the second pair, $2a + 6$, the HCF is 2. Factoring it out gives $2(a + 3)$.
- The expression is now $b(a + 3) + 2(a + 3)$.
- **Step 3:** The common bracket is $(a + 3)$. We factor this out.
- What's left over? The b and the $+2$. They go in the second bracket.
- **Final Answer:** $(a + 3)(b + 2)$.

5 Completing the Square

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1 The Two-Step Method for Completing the Square

Completing the square is a simple, two-step process for any quadratic expression $ax^2 + bx + c$.

The Core Process

1. **List Your Elements.** Before you start, list these four key values from your expression.

Elements List

$a = \dots \quad b = \dots \quad c = \dots \quad \frac{b}{2a} = \dots$

2. **Use the Standard Structure.** Memorize this structure. You will substitute your listed elements into it.

The Structure

$a(\dots)^2 - a(\dots)^2 + c$

Now, just fill in the blanks:

- The first bracket always contains **x**.
- The number inside **both** brackets is always the value of $\mathbf{\frac{b}{2a}}$ that you calculated.
- The first term, $a(x + \frac{b}{2a})^2$, stays as it is. You only need to use your calculator to compute the numerical part: $-a(\frac{b}{2a})^2 + c$.

Write $2x^2 + 8x + 5$ in completed square form.

1. **List Elements:**
   $a = 2 \quad b = 8 \quad c = 5 \quad \frac{b}{2a} = \frac{8}{2(2)} = 2.$
2. **Apply Structure:**
   $$a(x + \frac{b}{2a})^2 - a(\frac{b}{2a})^2 + c$$

$$\implies 2(x+2)^2 - 2(2)^2 + 5$$

3. **Simplify:** Keep the bracket part and calculate the rest.

$$2(x+2)^2 - 2(4) + 5$$

$$2(x+2)^2 - 8 + 5$$

Final Answer: $2(x+2)^2 - 3$.

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2 Application: Finding the Turning Point

After completing the square, you can find the vertex (turning point) of the parabola instantly. This point is either a **minimum** (if $a > 0$) or a **maximum** (if $a < 0$).

Finding the Turning Point from the Form $a(x-h)^2 + k$

The turning point is always at the coordinates $(\mathbf{h}, \mathbf{k})$.

- **The x-coordinate (h):** is the value inside the bracket, but with the **sign flipped**.
- **The y-coordinate (k):** is the number on the **outside**, with the **same sign**.

Find the turning point of the curve $y = 2(x+2)^2 - 3$.

- The number inside the bracket is $+2$. We flip the sign to get $\mathbf{-2}$.
- The number outside is $\mathbf{-3}$. We keep the sign.
- The turning point is $(-2, -3)$.
- Since $a = 2$ (which is positive), this point is a **minimum**.

Another example: Find the turning point of $y = -(x-3)^2 + 4$.

- The number inside is -3. Flip the sign to get $\mathbf{+3}$.
- The number outside is $\mathbf{+4}$. Keep the sign.
- The turning point is $(3, 4)$.
- Since $a = -1$ (which is negative), this point is a **maximum**.

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6 Simplifying Algebraic Fractions

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The Golden Rule of Algebraic Fractions

The first step for **ANY** algebraic fraction problem is to **FACTORISE** everything you can (numerators and denominators). This is the most important rule.

1 Multiplying and Dividing Fractions

Multiplication: Cancel First!

The rule is simple: **before you multiply, cancel**. You can cancel any common factor (bracket, letter, or number) from any numerator with any denominator.

- **Example:** To solve $\frac{x^2-9}{x+2} \times \frac{3x+6}{x-3}$:
  1. Factorise everything: $\frac{(x-3)(x+3)}{x+2} \times \frac{3(x+2)}{x-3}$.
  2. Cancel the common brackets: The $(x-3)$ on top cancels the $(x-3)$ on the bottom. The $(x+2)$ on the bottom cancels the $(x+2)$ on top.
  3. What's left? Just $x+3$ and 3.
  4. **Answer:** $3(x+3)$.

Division: Use the "KFC" Rule

Division is just multiplication in disguise. Use the **KFC** rule: **K**eep, **F**lip, **C**hange.

1. **Keep** the first fraction as it is.
2. **Flip** the second fraction upside down.
3. **Change** the division sign to a multiplication sign.
4. Now it's a multiplication problem, so you just follow the "Cancel First!" rule from above.

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2 Adding and Subtracting Fractions (The Critical Skill)

This is the most complex operation, but it's systematic. The goal is to create one big fraction.

How to Add or Subtract Fractions

1. **Create a single fraction bar and find the LCM.** The new denominator is the Lowest Common Multiple (LCM) of all the original denominators.

Finding the LCM

The LCM must contain **every different bracket, letter, and number** from all denominators.

- **Brackets/Letters:** You can see them. Just collect all the unique ones.
- **Numbers:** Find the LCM of the numbers (e.g., using prime factors if needed).

2. **Adjust the Numerators.** For each original fraction, look at its numerator. You must multiply it by whatever was "missing" from its original denominator to form the new LCM.
3. **Expand and Simplify the Numerator.** The problem now becomes an "Expand and Simplify" task for the new numerator. Be very careful with signs, especially if you are subtracting.

Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

1. **Find LCM:** The denominators are different brackets, so the LCM is $(x-2)(x+1)$.
2. **Adjust Numerators:**
   - The first numerator (4) needs to be multiplied by $(x+1)$.
   - The second numerator (3) needs to be multiplied by $(x-2)$.

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The expression becomes: $\dfrac{4(x+1)-3(x-2)}{(x-2)(x+1)}$.

3. **Expand and Simplify Numerator:**

$$4(x+1)-3(x-2)=4x+4-3x+6=x+10$$

Final Answer: $\dfrac{x+10}{(x-2)(x+1)}$.

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3 Mixed Operations (BODMAS)

Just like with normal numbers, when you see a mix of operations ($+, -, \times, \div$), you must follow the order of operations.

Order of Operations Rule

- **Priority:** Division ($\div$) and Multiplication ($\times$) must be done **FIRST**.
- **Secondary:** Addition ($+$) and Subtraction ($-$) are done **LAST**, using the LCM method.

Simplify: $\frac{x}{3} + \frac{2}{x} \div \frac{4}{x^2}$

1. **Step 1: Division First.** We must handle the $\div$ part before the $+$.

$$\frac{2}{x} \div \frac{4}{x^2} \to \frac{2}{x} \times \frac{x^2}{4} = \frac{2x^2}{4x} = \frac{x}{2}$$

2. **Step 2: Rewrite the Problem.** Now substitute this back.

$$\frac{x}{3} + \frac{x}{2}$$

3. **Step 3: Addition (LCM).** The LCM of 3 and 2 is 6.

$$\frac{x(2)}{6} + \frac{x(3)}{6} = \frac{2x + 3x}{6}$$

4. **Final Answer:** $\frac{5x}{6}$

4 Compound Fractions

This is when you have a "fraction within a fraction". These look scary, but there is a magic trick to solve them in one step.

The Magic Trick: Multiply All

Instead of trying to combine the small fractions, do this:

1. Identify the **denominator** of the small "inner" fraction.
2. Multiply **EVERY term** (top and bottom of the big fraction) by this denominator.
3. The fractions will disappear instantly!

Simplify: $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x^2}}$

1. **Identify Inner Denominators:** We have x and x^2. The largest one is x^2.
2. **The Trick:** Multiply **everything** (top and bottom) by x^2.

$$\frac{x^2(1)+x^2(\frac{1}{x})}{x^2(1)-x^2(\frac{1}{x^2})}$$

3. **Simplify:**
   - Top: x^2+x
   - Bottom: x^2-1

   Now the fraction is: $\dfrac{x^2+x}{x^2-1}$
4. **Factorise and Cancel (The Golden Rule):**

$$\frac{x(x+1)}{(x-1)(x+1)}$$

   Cancel the $(x+1)$ terms.
5. **Final Answer:** $\dfrac{x}{x-1}$

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Crucial Exam Advice

This topic is **VERY IMPORTANT**. Questions on algebraic fractions are a major part of algebra and frequently appear in exams. Direct problems can make up **at least 10% of your exam marks**. Mastering these methods is essential for a high grade.

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7 Rearrange Formulas

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1 The Universal Method for Rearranging Formulas

This method works for **ALL** cases, whether the subject you want appears once, twice, or many times. It's a reliable, step-by-step process.

Preliminary Step: Handling Powers and Roots

Before you begin, get rid of any overall square roots or powers.

- To remove a square root, **square both sides**.
- To remove a power (like ^{2}), **square root both sides**.

The Two-Stage Process

Stage 1: The Cleaning Step (Preparation)

Prepare the formula by removing any obstacles.

- **Expand Brackets:** If the subject is trapped inside a bracket, you must expand it to free it.
- **Eliminate Denominators:** If there are fractions, multiply all terms by the denominator (or the LCM) to clear them.

Stage 2: The Isolation Step (Solving)

Now that the equation is "clean", get the subject by itself using these three essential actions in order:

1. **Collect** Gather **all terms containing your subject** on one side, and move **everything else** to the other side.
2. **Factorise** This is the most important trick. If the subject is in more than one term, **take it as a common factor**. This forces the subject to appear only once.
3. **Divide** To get the subject completely alone, **divide** by whatever is multiplied by it (this is usually the bracket you just created).

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2 Worked Examples

Make x the subject of $A = \frac{9(1-4x)}{2g}$.

Cleaning (Fraction): Multiply both sides by $2g$.

$$2gA = 9(1 - 4x)$$

Cleaning (Bracket): Subject x is inside the bracket, so we must expand.

$$2gA = 9 - 36x$$

Isolation - Step 1 (Collect): Move the term with x $(-36x)$ to the left to make it positive. Move everything else to the right.

$$36x = 9 - 2gA$$

Isolation - Step 2 (Factorise): The subject x only appears in one term, so this step is not needed.

Isolation - Step 3 (Divide): Divide by 36.

$$x = \frac{9 - 2gA}{36}$$

Final Answer: $x = \dfrac{9 - 2gA}{36}$.

Make x the subject of the formula $p = \frac{2-ax}{x-b}$.

Cleaning (Fraction): Multiply both sides by the denominator $(x - b)$.

$$p(x - b) = 2 - ax$$

Cleaning (Bracket): Expand the left side to free the x.

$$px - pb = 2 - ax$$

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Isolation - Step 1 (Collect): Gather all terms with x on the left, and everything else on the right.

$$px + ax = 2 + pb$$

Isolation - Step 2 (Factorise): **This is the crucial step**. The subject x is in two terms, so we take it as a common factor.

$$x(p + a) = 2 + pb$$

Isolation - Step 3 (Divide): Divide by the bracket $(p + a)$ to get x alone.

$$x = \frac{2 + pb}{p + a}$$

Final Answer: $x = \dfrac{2 + pb}{p + a}$.

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8 Algebraic Proof

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1 The Language of Proof: Representing Numbers

The first step in any proof is to translate words into algebra. This table is your dictionary.

The Algebraic Dictionary

If the question says...	You write...
Any integer	n
Two different integers	n and m
Consecutive integers	n and $n+1$
An even number	$2n$
An odd number	$2n+1$
Consecutive even numbers	$2n$ and $2n+2$
Consecutive odd numbers	$2n+1$ and $2n+3$
A multiple of 'k' (e.g., multiple of 3)	kn (e.g., $3n$)
One more than a multiple of 'k'	$kn+1$
A square number	n^2

2 The Core Method: A Three-Step Process

All algebraic proofs follow the same basic logic: create an expression, simplify it, and then factorise it to show the property you need.

The Main Strategy

1. **Translate:** Use the table above to turn the words of the problem into an algebraic expression.
2. **Expand and Simplify:** Expand any brackets and collect all the like terms. Your

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goal is to get the simplest possible expression.

3. **Factorise to Prove:** This is the final and most important step. You manipulate the simplified expression to fit the form you need.

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3 Common Proof Scenarios and Their Tricks

Goal: Show the final expression can be written as $2 \times (\text{integer})$.
Trick: After simplifying, take **2** as a common factor.

Problem: Prove that $(n+3)^2 - (n-3)^2$ is always even.

- **Translate & Simplify:** $(n^2 + 6n + 9) - (n^2 - 6n + 9) = 12n$.
- **Factorise to Prove:** We need to show this is a multiple of 2.

$$12n = \mathbf{2}(6n)$$

- **Conclusion:** Since $6n$ is an integer, the expression is in the form $2 \times (\text{integer})$, so it is always even.

Goal: Show the final expression can be written as $2 \times (\text{integer}) + 1$.
Trick: After simplifying, take **2** as a common factor from the algebraic parts, leaving a $+1$ at the end.

Problem: Prove the difference between two consecutive square numbers is always odd.

- **Translate:** The numbers are n and $n+1$. The difference of squares is $(n+1)^2 - n^2$.
- **Simplify:** $(n^2 + 2n + 1) - n^2 = 2n + 1$.
- **Factorise to Prove:** We need the form $2 \times (\text{integer}) + 1$.

$$2n + 1 = \mathbf{2}(n) + 1$$

- **Conclusion:** Since n is an integer, the expression is in the form $2 \times (\text{integer}) + 1$, so it is always odd.

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Goal: Show the final expression can be written as $k \times (\text{integer})$.
Trick: After simplifying, take **k** as a common factor.

Problem: Prove that the difference between the squares of any two consecutive odd numbers is a multiple of 8.

- **Translate:** Consecutive odd numbers are $2n+1$ and $2n+3$. Difference of squares is $(2n+3)^2 - (2n+1)^2$.
- **Simplify:** $(4n^2 + 12n + 9) - (4n^2 + 4n + 1) = 8n + 8$.
- **Factorise to Prove:** We need to show it's a multiple of 8.

$$8n + 8 = \mathbf{8}(n+1)$$

- **Conclusion:** Since $n+1$ is an integer, the expression is in the form $8 \times (\text{integer})$, so it is always a multiple of 8.

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9 Solving Linear Equations

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1 The Universal Method for Solving Equations

Solving a linear equation is the same skill as rearranging a formula. The goal is to make the unknown (like x) the **subject**. The following two-stage process works for ALL linear equations.

The Main Strategy: Clean then Isolate

Stage 1: The Cleaning Step (Preparation)

Your first goal is to make the equation simple by removing obstacles.

- **Expand Brackets:** If there are any brackets, expand them to free the terms.
- **Eliminate Denominators:** If there are fractions, multiply every term in the equation by the Lowest Common Multiple (LCM) of the denominators. This cancels all fractions.

Stage 2: The Isolation Step (Solving)

After cleaning, the equation is simple. Now, isolate the unknown using these actions in order:

1. **Collect** Gather **all terms with your unknown** on one side, and move **everything else** to the other side. (Tip: Move the smaller 'x' term to avoid negatives).
2. **Add/Subtract** Combine the like terms on each side into single terms.
3. **Divide** **Divide** by the final number multiplied by your unknown to find its value.

2 Worked Examples

Solve the equation $5(x+3) = 3x - 4$.

Cleaning (Brackets): Expand the bracket on the left side.

$$5x + 15 = 3x - 4$$

Isolation (Collect): Move the smaller x-term ($3x$) to the left, and move the number ($+15$) to the right.

$$5x - 3x = -4 - 15$$

Isolation (Add/Subtract): Combine the terms on each side.

$$2x = -19$$

Isolation (Divide): Divide by 2.

$$x = \frac{-19}{2}$$

Final Answer: $x = -9.5$.

Solve the equation $\frac{4x-1}{5} + \frac{x+4}{2} = 3$.

Cleaning (Fractions): The denominators are 5 and 2. Their LCM is 10. Multiply **every term** by 10.

$$10\left(\frac{4x-1}{5}\right) + 10\left(\frac{x+4}{2}\right) = 10(3)$$

After cancelling, this simplifies to:

$$2(4x-1) + 5(x+4) = 30$$

Cleaning (Brackets): Now expand the brackets.

$$8x - 2 + 5x + 20 = 30$$

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Isolation (Collect & Add): Combine all the like terms on the left side.

$$13x + 18 = 30$$

Now, move the number (+18) to the right side.

$$13x = 30 - 18 \implies 13x = 12$$

Isolation (Divide): Divide by 13.

$$x = \frac{12}{13}$$

Final Answer: $x = \frac{12}{13}$.

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10 Solving Quadratics

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This Topic is Extremely Important!

Solving quadratic equations is one of the most critical skills in algebra. It is the foundation for solving non-linear simultaneous equations and appears in roughly **20% of all exam questions** in some form. Mastering this is not optional!

1 The Universal 2-Stage Method

Most quadratic equations don't start in the simple form. They are often complex. Your job is to follow this two-stage process.

The Main Strategy: Clean then Solve

Stage 1: The Cleanup Process

Your first goal is to get the equation into the **Standard Form**: $ax^2 + bx + c = 0$.

1. **Expand** any brackets.
2. **Rearrange** the equation so that one side is equal to zero.
3. **Collect Like Terms** to get a clean quadratic expression.

Stage 2: The Solving Process - Let The Calculator Guide You!

Once you have the standard form $ax^2 + bx + c = 0$, your calculator is your best friend.

Use your calculator's equation solver first!

The type of answer the calculator gives you tells you exactly which method to use for your working steps:

- **IF** the calculator gives **clean numbers** (integers or simple fractions)...
  $\longrightarrow$ Then you must solve by **Factorisation**. Your working must show the factors.

- **IF** a calculator não funcionar, experimente o seguinte: **messy decimals or surds**...
  $\longrightarrow$ Then you must solve using the **Quadratic Formula**. Your working must show the substitution into the formula.

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2 Worked Examples for Each Method

Method 1: Solving by Factorisation

Solve the equation $x^2 + 9x + 20 = 0$.

1 **Calculator Check** Inputting $a = 1, b = 9, c = 20$ into the calculator gives the clean answers $x = -4$ and $x = -5$. This tells us we must use factorisation.

2 **Find Factors** To get the factors for our working, we reverse the solutions:

- $x = -4 \implies (x + 4) = 0$
- $x = -5 \implies (x + 5) = 0$

3 **Show Working** Write the factorised equation.

$$(x + 4)(x + 5) = 0$$

4 **State Solutions** Final Answer: $x = -4$ or $x = -5$.

Method 2: The Quadratic Formula

Solve $3x^2 + 4x - 12 = 0$. Give your solutions to 2 d.p.

1 **Calculator Check** Inputting $a = 3, b = 4, c = -12$ gives messy decimals (1.44 and -2.77). This tells us we must use the formula.

2 **Write Formula and Substitute (CRUCIAL)**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(4) \pm \sqrt{(4)^2 - 4(3)(-12)}}{2(3)}$$

3 **State Solutions from Calculator** Final Answer: $x = 1.44$ or $x = -2.77$.

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Method 3: Completing the Square (Special Case)

Solve $(x-2)^2 = 3$. Give solutions to 3 s.f.

When to Use: The equation is already in "completed square" form. This is the only time you should choose this method (unless the question forces you).

1. **Isolate the Square** The squared bracket is already isolated.
2. **Square Root Both Sides** Don't forget the $\pm$ symbol!

$$x - 2 = \pm\sqrt{3}$$

3. **Solve for x** Add 2 to both sides.

$$x = 2 \pm \sqrt{3}$$

4. **Use Calculator for Final Answers** Final Answer: $x = 3.73$ or $x = 0.268$.

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11 Solving Inequalities

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1 Solving Linear Inequalities

This is identical to solving equations, with one special rule. We use the same two-stage method.

Universal Method

1. **Cleaning Step:** Prepare the formula by expanding brackets and eliminating denominators (by multiplying by the LCM).
2. **Isolation Step:** Isolate the unknown by **Collecting** terms with the subject on one side, then **Adding/Subtracting**, and finally **Dividing**.

The Golden Rule of Inequalities

There is only ONE difference from solving equations: **When you multiply or divide both sides by a negative number, you MUST flip the inequality sign!**

$$> \to < \qquad \geq \to \leq$$

Linear Inequality

Solve the inequality $5 - 2x \leq 21$.

Isolate: Subtract 5 from both sides: $-2x \leq 16$.

Divide by Negative: Now divide by -2. Because we are dividing by a negative, we **must flip the sign**.

$$\frac{-2x}{-2} \geq \frac{16}{-2}$$

Final Answer: $x \geq -8$.

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2 Solving Quadratic Inequalities

The Sketching Method

1. **Cleanup Step:** Rearrange to get the standard form $ax^2 + bx + c > 0$. Ensure the x^2 term is positive.
2. **Find Critical Values:** Treat as an equation ($ax^2 + bx + c = 0$) and solve with your calculator to find the roots.
3. **Sketch:** Draw a U-shaped parabola crossing the x-axis at your critical values.
4. **Identify Solution Region:**
   - For > 0 (Greater than), the solution is the two regions **outside** the roots ($x <$ small root OR $x >$ big root).
   - For < 0 (Less than), the solution is the single region **between** the roots (small root $< x <$ big root).

Quadratic Inequality

Solve $x^2 + 2x > 6x + 5$.

Cleanup: $x^2 - 4x - 5 > 0$.

Critical Values: Solving $x^2 - 4x - 5 = 0$ gives roots $x = 5$ and $x = -1$.

Sketch & Solve: The inequality is > 0 (Greater than), so we need the regions **outside** the roots.

Final Answer: $x < -1$ or $x > 5$.

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3 Compound Inequalities

This is a "three-sided" inequality like $-7 \leq 2x - 3 < 5$. The goal is to isolate x in the middle.

The "Do it to All Sides" Rule

Whatever operation you perform to isolate x, you must do it to the **left**, **middle**, and **right** parts of the inequality.

Compound Inequality

Solve $-7 \leq 2x - 3 < 5$.

1 **Add 3 to all sides** To get rid of the -3 in the middle:

$$-7 \boxed{\mathbf{+3}} \leq 2x \boxed{\mathbf{+3}} - 3 < 5 \boxed{\mathbf{+3}}$$

$$-4 \leq 2x < 8$$

2 **Divide all sides by 2** To isolate the x:

$$\frac{-4}{2} \leq \frac{2x}{2} < \frac{8}{2}$$

Final Answer: $\boxed{-2 \leq x < 4}$.

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4 Simultaneous Inequalities

This means finding the integer values that satisfy two *separate* inequalities at the same time.

The Number Line Overlap Method

1. **Solve Both Separately:** Solve each inequality on its own.
2. **Draw & Represent:** Draw a number line. Represent the first solution above the line and the second solution below it.
3. **Find the Overlap:** The final answer is the integer values in the region where the two solution lines overlap.

Simultaneous Inequalities

Find all integers n that satisfy $3n + 2 \leq 14$ and $n^2 - 6n + 5 < 0$.

Solve Both:

- *Inequality 1:* $3n \leq 12 \implies \mathbf{n \leq 4}$.
- *Inequality 2:* This is a quadratic. The calculator gives roots $n = 1$ and $n = 5$. Since it's < 0 (Less than), the solution is between the roots: $\mathbf{1 < n < 5}$.

Number Line Overlap: Draw a number line. Show $n \leq 4$ and $1 < n < 5$. The region where they overlap is from 1 up to 4, i.e., $1 < n \leq 4$.

List Integers: The integers in the region $1 < n \leq 4$ are 2, 3, and 4.

Final Answer: $n = 2, 3, 4$.

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12 Simultaneous Equations

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1 Linear Simultaneous Equations

The Main Strategy

1. Get the final answer first using your calculator. This helps you check your work.
2. Multiply (if needed): Multiply one or both equations so that the coefficient of either x or y is the same number in both.
3. Add or Subtract:
   - If the signs of the matching coefficients are **different**, **ADD** the equations.
   - If the signs are the **same**, **SUBTRACT** the equations.
4. Solve the resulting simple equation to find the first variable.
5. Substitute this value back into one of the original equations to find the second variable.

Problem: Solve: $5x + 2y = 11$ and $4x - 3y = 18$.

Step 1: Multiply We'll eliminate y. Multiply the first eq. by 3 and the second by 2.

$$(5x + 2y = 11) \times 3 \implies 15x + 6y = 33$$
$$(4x - 3y = 18) \times 2 \implies 8x - 6y = 36$$

Step 2: Add The signs for $6y$ are different, so we **ADD** the two new equations.

$$(15x + 8x) + (6y - 6y) = (33 + 36)$$
$$23x = 69$$

Step 3: Solve Divide by 23:

KEY FORMULA

$x = 3$

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Step 4: Substitute Put $x = 3$ into the first original equation:

$$\begin{aligned} 5(3) + 2y &= 11 \\ 15 + 2y &= 11 \\ 2y &= -4 \end{aligned}$$

KEY FORMULA

$y = -2$

Final Answer

Solution: $(x = 3, y = -2)$

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2 Non-Linear (Quadratic) Simultaneous Equations

This is a HUGE Topic!

This is one of the most important and frequently tested topics!

You know you have this type of problem if you see keywords like **intersect, meet, solutions**, or if you are given **two unknowns** with one equation having a power of 2 **(like x^2, y^2, xy)**.

The 5-Step Method for Non-Linear Equations

1. Rearrange the Linear: From the linear equation, make x or y the subject. **Choose the variable with a coefficient of 1 to avoid fractions.**
2. Substitute: Substitute the expression from Step 1 into the quadratic equation.
3. Cleanup Process: Your new equation is a complex quadratic. You must now **Expand, Rearrange, and Collect** to get it into the standard form

   KEY FORMULA

   $ax^2 + bx + c = 0$

4. Solve: Solve the new quadratic equation using your calculator (Factorise or Formula method). This will give you two values for one variable.
5. Find the Other Variable: Substitute your two answers from Step 4 back into the **rearranged linear equation from Step 1** to find the corresponding two values of the other variable.

Problem: Solve: $x^2 + y^2 = 36$ and $x - 2y = 6$.

Step 1: Rearrange Linear From $x - 2y = 6$, it is easiest to make x the subject:

KEY FORMULA

$x = 2y + 6$

Step 2: Substitute Substitute this into the quadratic equation:

$$(2y+6)^2 + y^2 = 36$$

Step 3: Cleanup Process

- **Expand:** $(4y^2 + 24y + 36) + y^2 = 36$
- **Rearrange & Collect:** $5y^2 + 24y + 36 - 36 = 0 \implies \mathbf{5y^2 + 24y = 0}$

Step 4: Solve Factorise the new quadratic: $y(5y+24) = 0$. This gives two solutions:

$$y = 0$$
$$y = -\frac{24}{5}$$

Step 5: Find Other Variable Substitute these y-values back into $x = 2y + 6$:

- If $y = 0$, then $x = 2(0) + 6 = 6$
- If $y = -24/5$, then $x = 2(-24/5) + 6 = -48/5 + 30/5 = -18/5$

Final Answer

Paired Solutions: $(x = 6, y = 0)$ and $\left(x = -\frac{18}{5}, y = -\frac{24}{5}\right)$

Pro Tip

Always double-check your solutions by substituting!

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13 Forming and Solving Equations

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1 Forming Equations from Words

4-Step Method for Word Problems

1. **Identify the unknown**: Choose a variable (e.g., x) for the unknown quantity.
2. **Translate to algebra**: Convert word phrases into mathematical expressions using the guide below.
3. **Form the equation**: Look for key phrases like "is equal to", "total is", or "difference is" to link the expressions.
4. **Solve and contextualize**: Solve the equation and write the final answer in the context of the original problem.

Translation Guide: Words to Algebra

Operation	Key Words	Algebra	Example
Addition	sum, total, more than, increase	$+$	"5 more than x" $\to x+5$
Subtraction	difference, less than, decrease	$-$	"3 less than y" $\to y-3$
Multiplication	product, times, double, triple	$\times$	"twice x" $\to 2x$
Division	shared, split, halved, quartered	$\div$	"half of x" $\to \frac{x}{2}$

Worked Example: Flower Problem

Problem: Yellow flowers are triple the red flowers. Purple flowers are 5 more than yellow. The difference between purple and red is 29. Find the number of yellow flowers.

Step 1 (Define) Let red flowers = x.

Step 2 (Translate) Yellow = $3x$, Purple = $3x+5$.

Step 3 (Form Eq.) Difference: $(3x+5)-x=29$.

Step 4 (Solve) $2x+5=29 \implies 2x=24 \implies x=12$.

Final Answer

Number of yellow flowers $= 3x = 3(12) = \mathbf{36}$.

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2 Forming Equations from Shapes

3-Step Method for Shape Problems

1. **Label the Diagram**: Write expressions for all unknown lengths/angles.
2. **Use Geometric Properties**: Apply formulas for perimeter, area, angles, etc.
3. **Set Up and Solve**: Equate expressions to given values and solve.

Worked Example: Rectangle Problem

Problem: A rectangle has length $3x+1$ and width $2x-5$. Its perimeter is 22 cm. Find the area.

Property Perimeter of a rectangle.

Rule $P = 2(L+W)$.

Form Equation $22 = 2((3x+1)+(2x-5))$.

Solve for x

$$22 = 2(5x-4)$$
$$11 = 5x-4 \implies 15 = 5x \implies \mathbf{x = 3}$$

Find Area Length $= 3(3)+1 = 10$. Width $= 2(3)-5 = 1$. Area $= 10 \times 1 = 10$.

Final Answer

The area is **10** cm^2.

3 Problem Solving with Equations

The Universal Problem-Solving Checklist

This approach applies to almost any complex problem that requires you to build and solve an equation.

1. **Form the Equation:** Use the context (words, shapes, simultaneous relationships) to build the algebraic equation(s).
2. **Clean Up:** **Expand**, **Rearrange**, and **Collect** to get the equation into its simplest standard form (e.g., $ax^2 + bx + c = 0$).
3. **Solve:** Solve the final equation using the appropriate method.
4. **Check the Context:** Does the answer make sense? (e.g., length cannot be negative). Make sure you answer the specific question asked.

Worked Example: Trapezium Area

Problem: The area of a trapezium (parallel sides $x - 4, x + 5$; height $2x$) is 351 cm^2. Show $2x^2 + x - 351 = 0$, and find x.

Form Equation:

$$351 = \frac{1}{2}((x-4) + (x+5)) \times (2x)$$

Clean Up:

$$\begin{aligned} 351 &= \frac{1}{2}(2x+1)(2x) \\ 351 &= (2x+1)(x) \\ 351 &= 2x^2 + x \\ 0 &= 2x^2 + x - 351 \end{aligned}$$

Solve: Using calculator: $x = 13$ and $x = -13.5$.

Check Context: Length cannot be negative, so reject $x = -13.5$.

Final Answer

The value of x is **13**.

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